Flower Image Classification

Final report - frozen selection and held-out test evaluation

1. Dataset and experimental methods

Flower Photos contains five folder labels: daisy, dandelion, roses, sunflowers, and tulips. From 3,670 original images, the audit excluded two redundant sunflower copies and both copies of one conflicting-label image. The 3,666 accepted images use fixed, stratified seed-42 splits: 2,564 train, 551 validation, and 551 test. All model selection uses validation macro-F1, with lower validation loss as the tie-breaker. The frozen model list and hashes precede test inference.

Class	Train	Validation	Test
daisy	443	95	95
dandelion	628	135	135
roses	448	96	96
sunflowers	487	105	105
tulips	558	120	120
Total	2564	551	551

The custom CNN has three convolution/BatchNorm/ReLU/pooling blocks and a five-class classifier. Its baseline and augmented runs each used 12 epochs from the same initial weights. ResNet18 first trained a five-class head with its backbone and BatchNorm statistics frozen for six epochs, then fine-tuned all layers for six epochs. Day 5 compares two matched continuations from the same best original fine-tuned weights. Both use AdamW, learning rate 0.00001, weight decay 0.0001, cosine scheduling, and the same seed/budget. Only training crop area changes: (0.75, 1.0) versus (0.5, 1.0).

Inputs use RGB/EXIF correction and 224-pixel model inputs. Training augmentation includes random crop, horizontal flip, 15-degree rotation, and mild color jitter. Validation, test, and demo inputs use Resize(256), CenterCrop(224), and ImageNet normalization. Test scores do not change the selected model. Per-image predictions, per-class reports, confusion matrices, histories, and runtime versions accompany this report.

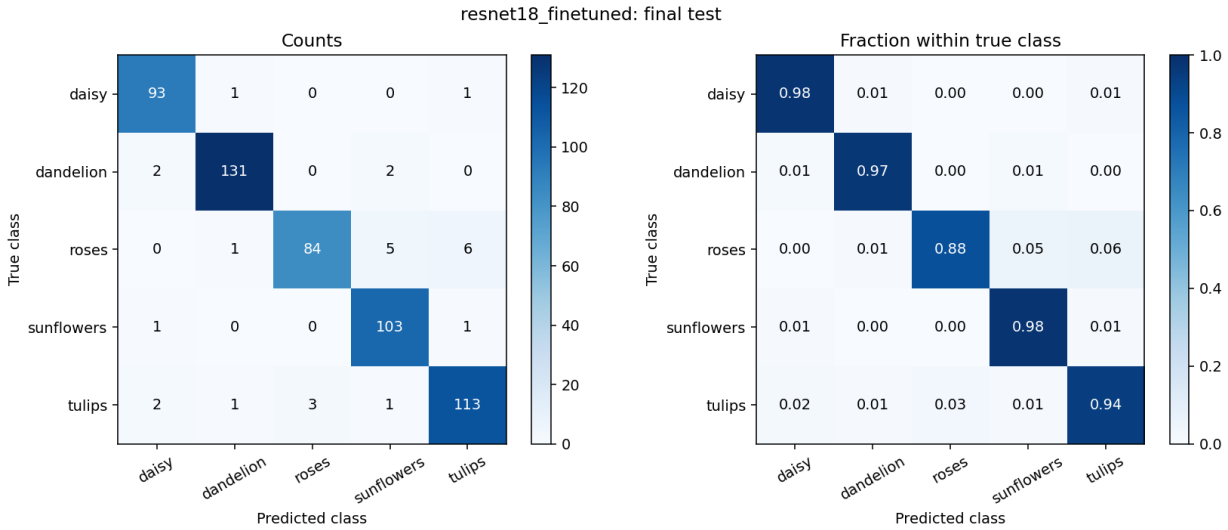
Split manifest SHA-256: 4fc79e62cb179ff75a18772ba16f72e091184263e6fb8f4731819bbdc07c2dbb

2. Results and the controlled crop experiment

Model	Val acc.	Val F1	Test acc.	Test F1	Epoch budget
CNN baseline	68.06%	0.6740	73.32%	0.7293	12
CNN augmented	67.51%	0.6667	72.78%	0.7207	12
ResNet head	84.75%	0.8460	89.47%	0.8928	6
ResNet finetuned	94.01%	0.9393	95.10%	0.9494	12
ResNet control	93.83%	0.9379	95.46%	0.9530	18
ResNet crop	93.65%	0.9357	96.01%	0.9577	18

The crop arm's validation macro-F1 was 0.9357, versus 0.9379 for matched continued training. It was lower by 0.0022. The final validation-selected model is resnet18_finetuned.

Selected model: resnet18_finetuned. Validation accuracy 94.01%, macro-F1 0.9393. Final test accuracy 95.10%, macro-F1 0.9494, on 551 test images. The selected model remained fixed after testing.



Selected model confusion matrix. Rows are true classes; columns are predictions. The right panel normalizes within each true class.

The four earlier models retain their original validation-selected checkpoints. Day 5's matched control isolates additional training from the crop intervention. Negative or unchanged results are reported without claiming improvement.

3. Error analysis, demo, and limitations

Day 4's largest validation confusions were dandelion to daisy (7), roses to tulips (7), and tulips to roses (5). Roses had the lowest per-class F1. The 40-case diagnostic review covered 20 correct and 20 incorrect predictions. Review attribution: 13 user and 27 authorized AI-assisted visual reviews. Error tags included six similar-class cases, five clutter cases, and three small/distant subjects. Heatmaps sometimes highlighted surrounding objects. These observations motivate a hypothesis, not a causal conclusion.



Validation example val_0035: a reviewed small/distant-subject error. Panels show the exact input crop, predicted-class Grad-CAM, and true-class Grad-CAM. Heatmaps normalize independently.

One seed and one split limit generalization claims. ResNet18 benefits from ImageNet pretraining and greater capacity. The head-only run had six epochs, original fine-tuning had 12 allocated epochs including head training, and each Day 5 arm adds 6 epochs from the same selected fine-tuned checkpoint. Budgets differ across the six models. The review sample favors high-confidence errors and is not representative of all failures. Exact pixel duplicate checks do not eliminate near duplicates or every label problem. Grad-CAM is approximate. Scores are uncalibrated. Smaller training crops can exclude flowers.

The upload demo uses the frozen validation-selected checkpoint and the same evaluation preprocessing. It returns five class scores and a predicted-class heatmap. A private notebook form and a localhost Gradio app are included. The classifier always assigns one of the five labels, even to unfamiliar images.

Sources: TensorFlow Flower Photos image-loading tutorial (tensorflow.org/tutorials/load_data/images); PyTorch transfer-learning tutorial (docs.pytorch.org/tutorials/beginner/transfer_learning_tutorial.html); Selvaraju et al., Grad-CAM, ICCV 2017. Individual image credits are preserved in docs/DATASET_IMAGE_CREDITS.txt in the backup.